

The mayfly (Insecta, Ephemeroptera) taxocoene of small intermittent stream – a Central European case study

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Abstract

In Central Europe the effects of anomalous discharges on macroinvertebrate assemblages have been studied only in perspective of management of artificial discharges. However, recurrent drying-out of small streams in last decade showed the necessity to pay the attention to the impacts linked with global climatic changes. Therefore the objectives of this study were to describe the impact of drying-out on the composition of the mayfly taxocoene and to evaluate relevance of individual species traits for resisting of mayflies to drying-out. The mayfly taxocoene of two model localities, intermittent and permanent brooks, were investigated in 2005. Quantitative data from intensive sampling on intermittent stream showed a high fluctuation of total density depending on species life cycles and varying hydrological conditions. Comparing with the permanent stream, the mayfly taxocoene was deprived of rheobionts and high oxygen demand species, which meant considerable impoverishment of the taxocoene. To explain differences between both stream types in species survival and re-colonization ability, 25 species traits were evaluated. These included so-called “ecological traits” (e.g. habitat and substrate range, density, distribution, current speed adaptation, eurythermy) and “biological traits” connected with life cycle, adult stage, and larval stage. Only species showing the highest number of advantageous and minimum number of disadvantageous traits were able to successfully survive under unfavourable condition of intermittent brook. “Biological traits”, which were considered to be the most important, could be good predictors for assessing the sensitivity of species to extreme changes of temperature and hydrological regimes and species ability to re-colonize streams exposed to these impacts.

Key words

intermittent stream, Ephemeroptera, drought, life cycle

Introduction

Climate has globally changed many times over the course of the history of Earth. At present we are in a warm interglacial period, but over the past two million years, there have been numerous Ice Ages however the last one (Würm or Vistula Glaciation) seems to be more important from impact of climatic changes on biota point of view and evaluation of this process in general. Moreover, this period exerted a crucial impact on selection and composition of contemporary faunas (and biota in general), phylogeny and biodiversity of most animal groups and numerous processes at population of biosystem levels. The geological records shows that past climate changes has not always been gradual, rather there have been numerous changes that were very rapid, sometimes over period as short as centuries or even decades (Bonada et al., 2007), principal changes can be traced even within modern human's history ("New Age"). For instance, beginning around 1550 A.D. and lasting until around 1850 was a period of cold temperature, sometimes call the "Little Ice Age" (Lamb, 1995).

There are several future climate scenarios for individual regions of the world. Taking into consideration Europe, there are several alternatives which identically expect local and/or regional temperature increase and more variable precipitation patterns. In Central Europe no substantial changes of annual precipitation are anticipated, unlike the seasonal precipitation pattern. Winter-spring precipitation would increase and summer-autumn precipitation would decrease (Bonada et al., 2007). As far as temperature is concerned, temperatures in Europe warmed by 0.5 °C between the mid 1800s and 1940 but cooled again over the next 25 years. Since then temperatures have been rising and current predictions indicate increases of annual mean air temperatures as much as 8 °C by the end of century in some regions (cf. e.g., IPCC 2001)

Besides temperature increase and other impacts on the biosphere, climatic changes impact directly the hydrosphere. Naturally, the impact of climate changes on hydrosphere is substantially different than on the "rest" of biosphere, mainly terrestrial ecosystems and the effect might be similar and simultaneous as well as synergic or antagonistic. Therefore growing interest in detailed research of its particular impacted components is understandable. Considerable attention is paid to fluvial systems, which might be affected by the changes of hydrologic regimes, primarily in term of extreme hydrological events, such as high discharge in cold seasons and drying out in warm seasons. Focusing on the past decade in Central Europe, beside the phenomenon of summer drought (very unusual event in this region), there are also rising occurrences of summer floods. For instance, extreme floods occurred in the

Czech part of the Danube basin in July, 1997 and in Elbe basin in August 2002. On the other hand, the summer in 2003 was extremely hot and dry in Central Europe.

Apart from the problem of surplus or lack of water, it is necessary to be interested in the influences of anomalous discharges to aquatic fauna from the practical consequences point of view. Changes of the taxonomical composition and especially functional structure of assemblages could have far-reaching impacts on self-cleaning processes of water (and consequently also on oxygenic regime), as well as the productivity of freshwater biotopes. The impact of extremely low discharges, or drying out of various freshwater habitats, was frequent topic of researches in regions with common occurrence of these events, e.g. Australia (e.g. Brock et al., 2003; McMahon and Finlayson, 2003; review in Boulton, 2003), South and North America (Miller and Golladay, 1996; Del Rosario and Resh, 2000; Smith et al., 2001; Covich et al., 2003). In Europe drying out of streams were studied especially in the south of the continent (e.g. Pires et al., 2000), focusing on differences of taxonomical and functional structure of benthic assemblages between streams of temperate region and mediterranean region incl. temporary streams (Bonada et al. 2007). The Czech Republic, situated just in the centre of the European continent and at the boundaries of three sea-drainage areas (North Sea, Baltic Sea, and Black Sea), shows a high percentage of small (first to fourth order by Strahler) running waters represent approximately 90 % of the total length of the Czech rivers and streams. In the Czech Republic there are several regions, where small perennial streams, as well as intermittent streams, were found also in past. These regions are defined by specific geological conditions (karstic, e.g. the Moravian Karst and cretaceous regions, e.g. parts of the Czech Plateau) or it is heavily altered areas (deforested or meliorated areas, e.g. Pannonian lowlands in the south of the Czech Republic). And just these distinctive regions were impacted the most by the effects of extreme drought and above-normal summer temperatures in 2003. In this climatically extreme period (2002-2003), studies concerned on the assessment of the ecological estate of two nearby brooks in the boundary of Central Highlands and Pannonian lowlands (Illies, 1967) were realized. One of these brooks dried out, unlike the other. The results of the research conducted in intermittent brook (the Gránický Brook) were published (Řezníčková et al., 2007), whereas the data on the second, permanent, brook (the Klaperův Brook) were left unpublished in diploma thesis (Nyklová, 2006). Unpublished data from the permanent brook, which is in general similar in abiotic characteristics to the intermittent Gránický Brook, were used for the definition of the reference estate in this study. The comparison of these results stimulated follow-up research, which took place in 2005 in intermittent brook (the Gránický Brook). The results are being

processed (Řezníčková et al., 2007). In comparison with permanent brook, the intermittent brook showed evident and stable differences in the composition and species richness of benthic assemblages, which are not interpretable by the impacts of pollution or hydromorphologic dissimilarities of both brooks. Thus drying out seems to be the only disturbance. Surviving of dry period is dependent also on the characteristics of species, their biological and ecological traits. Species traits important for overcoming of this type of disturbance have not been summarized yet and there is also lack of detailed information on particular species. Therefore the objectives of this study were to summarize the information on species traits of species recorded on both sites studied, to evaluate relevance of individual traits for resisting to drying out and to verify the results on the comparison of taxocoenes of permanent and intermittent brooks. The mayflies (Ephemeroptera) have been chosen as a suitable model group due to their relatively easy identification even in early larval instars, relatively very low vagility and conservative distribution. The Ephemeroptera species involve in general (as well as in taxocoenes studied here) both less sensitive “generalists” with beneficial species traits as well as “specialists” with strict environmental limits and extensive knowledge on their life cycles and species traits.

Study area

Both streams, one permanent (the Klaperův Brook) and one with intermittent stretch (the Gránický Brook) are 3rd order tributaries of the Dyje (Thaya) River (the Danube catchment, (Fig. 1) situated in the warmest (mean annual air temperature is 8-9 °C) and driest area (sum of annual precipitation is 450-500 mm) of the Czech Republic. The presence of sandy clay sediments and schistose biotitic granites with high concentrations of sulphates is the main reason of higher values of conductivity of water, typical for the region.

The total length and catchment area of both streams is comparable (the Klaperův Brook: 8.5 km and 17.64 km², the Gránický Brook 13 km, and 20.5 km², respectively). The brooks have similar substrate (diverse, but cobbles dominating, boulder are rare). The distance of study sites, situated in the middle stretches of streams, is 10.5 km. The discharge of both brooks is comparable (ca 0.2 – 0.3 m³.s⁻¹) during winter and spring season. While the Klaperův Brook has permanent water discharge from spring to mouth, the upper part of the Gránický Brook has permanent character, middle part, where study site is located, is intermittent and its lower part is fed by groundwater and has perennial character with drought events only in extremely dry years. The intermittent stretch became completely dry every summer during last 12 years (1996 – 2008).

Material and Methods

The detail study of the Gránický Brook has started in early spring 2005 and finished in late autumn. Eleven series of samples were taken between April 5 and November 11 in about 3 weeks intervals (Fig. 2), both semiquantitative and quantitative methods sampling methods were used. Physico-chemical parameters (pH, conductivity, dissolved oxygen - DO, water temperature) and discharge were measured (Tab. 1).

During the first period of research (April 2002 do May 2003) the benthic macroinvertebrates were sampled in 6 weeks intervals (except winter) by water net at both brooks. Semiquantitative multihabitat sampling method (CSN 75 7701) was used. The material collected has been fixed in 4% formaldehyde in the field and transferred into 75 % alcohol after determination. It is deposited in collection of the Institute of Botany and Zoology, Faculty of Science, Masaryk University, Czech Republic.

Species traits

Selected species traits of mayfly species found in the Gránický Brook and/or in the Klaperův Brook were compiled from literature sources and original data. The five-grade scale were used to assess if the trait is advantageous for survival of drought events or not (if possible to distinguish with species studied): ++++ : most advantageous; +++ : highly advantageous or very pronounced; ++ : very advantageous or pronounced; + : advantageous or less pronounced; - : disadvantageous or not pronounced; (?): supposed, not exactly known; (??): literature data highly controversial. The scale +++ → – was defined only within the species collected). The following traits were taken into account: life cycle flexibility, bi- or polyvoltinism, fecundity, adaptation in oviposition, length of embryogenesis egg quiescence, egg survival of drought, degree of parthenogenesis, asynchrony in egg hatch, asynchronic development, asynchrony in emergence, length of winged stage, flight ability, passive transport, body form and length, larval growth rate, eurythermy, respiratory adaptation, feeding type, current speed adaptation, habitat range, substrate range, downstream drift, density and distribution. Score of most (4+), highly (3+) and very (2+) advantageous species trait and total score, i.e. No. of (+) and (–), were calculated.

The life cycles represent the most important species traits concerning drought impacts. There are several systems of classification and nomenclature of mayfly life cycles. While the classification by Clifford (1982: Quest. Entomol. 18: 15-90) seems to be most frequently used in general (cf., e.g. Studemann et al., 1994; Haybach, 1998, 2006; Sartori & Landolt, 1998; Bauernfeind & Humpesch, 2001; Derka 2003c), Central European species are often classified

according to Landa (1968, 1969) or Sowa (1975b). We have primarily used the Clifford's (1982) classification evaluating the life cycle type as follows (only categories found in species dealt with in this study are mentioned). Main life cycle categories U – Seasonal univoltine; MB – Seasonal bivoltine, MP – Seasonal polyvoltine; U-MB – Total uni-multivoltine; Y – Total semivoltine; U-Y – Total uni-semivoltine. Life cycle groups (or “subcategories”): Uw – Seasonal univoltine (winter); Us – Seasonal univoltine (summer); Uw-Us – Seasonal univoltine (winter-summer); MBss – Seasonal bivoltine (summer); MBws – Seasonal bivoltine (summer-winter); MB-MP – Seasonal bivoltine or polyvoltine; Us-MBss – Seasonal variable (1 or 2 summer generations); Uw-MBws – Seasonal variable (univoltine winter, possible summer generation); 2Y – Seasonal semivoltine (1 generation in 2 years); 2Y-3Y – Seasonal semivoltine (1 generation in 1 or 2 years); Uw-2Y – Seasonal variable (univoltine winter or two year semivoltine). However, since the two classification mentioned above (Landa, 1968; Sowa 1975b) are often used, we employed them, too, and consequently, we consider necessary to briefly present the equivalency among individual life cycle categories of these 3 classifications most frequently used. In other words, to define which Clifford's (1982) categories agree at least partially with those by Landa (1968) and/or Sowa (1975b).

Life cycle subcategories according to the classification by Landa (1968: *Acta Entomol. Bohemoslov.* **65**, 5: 276-284) [main life cycle types A – Seasonal univoltine (winter or summer); B – Seasonal univoltine (winter-summer) and/or seasonal bivoltine and variable; C – Seasonal semivoltine (1 generation in 2 or 3 years); D – Seasonal semivoltine (1 generation in 1 or 2 years) and/or seasonal variable (univoltine winter or two-year semivoltine)] shows the following approximative relationships to subcategories by Clifford (1982): A1, A3 = Uw; A2 = Us; B1 = MBws; B2 = MBss and/or Us-MBss; B3 = MB-MP and/or Uw-MBws; B4 = Uw-Us; C1 = 2Y; C2 = 3Y; D1 = Uw-2Y; D2 = 2Y-3Y (here simplified, see Soldán & Zahrádková, 2000 for further details).

Main life cycle types according to the classification by Sowa (1975b: *Acta Hydrobiol., Kraków* **17**, 4: 319-353) [main life cycle types: A – Seasonal semivoltine and/or seasonal variable (total uni-multivoltine), B – Seasonal univoltine (winter, summer, winter-summer); C – Seasonal bivoltine and/or polyvoltine and/or seasonal variable variable (1 or 2 summer generations). Life cycle groups (subcategories) exhibits the following approximative relationships to subcategories by Clifford (1982): A1 = Uw-MBws and/or 2Y; A2 = Uw-2Y and/or 2Y-3Y; B1 =Us; B2 and/or B3 = Uw; B4 and/or B5 = Uw-Us; C1 = MBss; C2 = MBws; C3 = MB-MP (here simplified, see Soldán & Zahrádková, 2000 for further details).

Results and discussion

Species composition of the mayfly taxocoene

Altogether 13 mayfly species of 9 genera belonging to 5 families have been found at the area studied (Tab. 2). Four species, namely *S. aestivalis*, *B. rhodani*, *Electrogena* sp. cf. *ujhelyii* (see below), and *H. fusca* form the Ephemeroptera taxocene of the Gránický Brook, all species mentioned below except for *S. aestivalis* belongs to mayfly taxocene of the Klaperův Brook only (see also Tab. 2 and 3)

Except for a single species of the genus *Electrogena* of the family Heptageniidae (subfamily Heptageniinae), species collected at both sites of the area studied (i.e. including also a “reference” locality, the Klaperův Brook) have been identified without any problems using current determination keys of Central European mayfly fauna (Landa 1969, Bauernfeind 1994a, 1994b, Bauernfeind and Humpesch 2001, Studemann 1992). The species identified and their taxonomic position are, as follows (delimitation and sequence of families and subfamilies after the classification by McCafferty 1991, genera, subgenera and species in the alphabetical order): *Siphonurus* (*Siphonurus* s.str.) *aestivalis* Eaton, 1903 of the Siphonuridae (subfamily Siphonurinae), *Baetis* (*Nigrobaetis*) *muticus* (Linnaeus, 1758) and *Baetis* (*Rhodobaetis*) *rhodani* (Pictet, 1843) of the Baetidae (subfamily Baetinae), *Ecdyonurus* (*Ecdyonurus* s.str.) *starmachi* Sowa, 1971, *Rhithrogena* (*Rhithrogena* s.str.) *carpatoalpina* Kłonowska, Olechowska, Sartori et Weichselbaumer, 1987 and *Rhithrogena* (*Rhithrogena* s.str.) *semicolorata* (Curtis, 1834) of the Heptageniidae (subfamily Heptageniinae), *Paraleptophlebia submarginata* (Stephens, 1836) and *Paraleptophlebia werneri* Ulmer, 1919 (Leptophlebiidae, Leptophlebiinae), *Habroleptoides confusa* Sartori & Jacob, 1986, *Habrophlebia* (*Habrophlebia* s.str.) *fusca* (Curtis, 1834), and *Habrophlebia* (*Habrophlebia* s.str.) *lauta* Eaton, 1884 (Leptophlebiidae, Habrophlebiinae), and *Ephemerella* (*Ephemerella* s.str.) *danica* Müller, 1764 of the Ephemeridae (subfamily Ephemerinae).

Concerning the only *Electrogena* species, there is a little doubt as to its proper taxonomic position. Judging from the arrangement of larval morphological characters (mouthparts, leg and posterior margin of abdominal terga chaetotaxy, gills and cerci) and exochorionic structure of eggs, this material most probably belongs to *Electrogena samalorum* (Landa, 1982), originally described sub *Ecdyonurus samalorum* Landa [in Landa & Soldán], 1982: Acta Entomol. Bohemoslov. 79: 31. Landa & Soldán 1982) also summarized critical distinguishing characters of this species from closely related *Electrogena ujhelyii* (Sowa, 1981), originally described sub *Ecdyonurus ujhelyii* Sowa, 1981; Acta Hydrobiol. (Kraków)

23: 376. Anyway, *E. samalorum* was found conspecific and synonymized with *E. ujhelyii* by Zurwerra & Tomka (1986), however the type material has never been compared. Moreover, *E. ujhelyii*, originally misidentified as *Ecdyonurus subalpinus* Klapálek, 1907 (by Ujhelyi 1966, see Sowa 1981: 375) has been later confused by the same authors (Tomka & Zurwerra 1985: 101) with related species *Electrogena gridellii* (Grandi, 1953) and *E. quadrilineata* (Landa, 1969) and described once again from Switzerland (sub *Electrogena rivuscellana* Sartori & Landolt [in Landolt et al.], 1991: Bull. Soc. Vaud. Sci. Nat. 80, 4: 459). The opinion of conspecificity of *E. ujhelyii* and *E. samalorum* is followed by some authors (e.g. Bauernfeind & Humpesch 2001), while the others (e.g., Derka 2003a, 2003b, 2003c in populations living in Slovakia) recognize validity of *E. samalorum*. Since *E. samalorum* still has not been formally removed from synonymy with *E. ujhelyii* and its proper taxonomic position thus remains questionable, we present our material like *Electrogena* sp. cf. *ujhelyii*.

Gránický Brook

Total density fluctuation

The Ephemeroptera taxocoene consisted of 4 species – *Siphonurus aestivalis*, *Baetis rhodani*, *Electrogena* sp. cf. *ujhelyii*, and *Habrophlebia fusca*.

The total density (see Fig. 3) fluctuated during the sampling season depending on (i) the life cycles of the species and (ii) hydrological events. Low total species density in April was a consequence of high discharge caused by rapid snowmelt at the end of March and the beginning of April.

Fluctuation of total density and especially the absence of *S. aestivalis* and *Electrogena* sp. cf. *ujhelyii* is strongly dependent, first of all, on their life cycles. Both represent seasonal univoltine winter species that have finished their development before drying-up and this process apparently exerts no effects on their numbers in samples. On the other hand drying up affected the second (summer) generation of *B. rhodani*; a seasonal bivoltine winter species. Partial effects of drying-up process were observed also in the seasonal univoltine winter species *H. fusca*, they occurred in the larval stage usually till July.

Habitat preferences of mayflies

Siphonurus aestivalis, preferred sites that exhibited, during larval development of the species, sufficient depths with slow flow. Younger larvae occurred predominantly at shallow, near bank gravel bar with leaves debris.

Baetis rhodani commonly occurred in large numbers in riffles; gradual decrease of water level caused the concentration of larvae at places with slower flow. The drying-out partially

affected the second (summer) generation. *B. rhodani* preferred habitats without fine substrate with non-zero bottom current velocity. Younger larvae occurred at habitats with CPOM.

Electrogena sp. cf. *ujhelyii*, did not exhibit any apparent affinity to habitats *per se*. Small and mid-sized larvae (body length of 2 – 8 mm) of *Electrogena* sp cf. *ujhelyii* preferred places with gravel and CPOM substrate, older larvae (> 8 mm) preferred places with pebbles and cobbles.

Habrophlebia fusca, showed the lowest densities in riffles preferring deeper places, namely those with character of pools. Shortly before drying out, larvae moved to microhabitats with shallow but stable water depth. *H. fusca* preferred shallow habitats with gravel-sand substrate with xylal. Partial effects of drying-up process were observed – the larvae usually occur till the end of July.

Life strategies and species traits in species collected

This contribution deals with mayflies (while other benthic biota will be treated later), among others, the reasons are as follows. (i) First of all, in Central Europe with absolutely prevailing permanent water courses, data describing the effects of drought, on one hand and foods on the other (more precisely, unfavourable predict scenarios not only in precipitation and runoff but also in temperature fluctuation) are still relatively very scarce or fragmentary; (ii) Mayflies, first appearing in the fossil records of the Upper Carboniferous in excess of 250 million years ago represent, with their only about 3,000 described species within 39 families a group in an evident regression with “old” and relatively firmly fixed adaptations especially in larval stage; (iii) Due to a relatively very low vagility, their distribution is conservative with a very high degree of endemism (but not in the area studied and Europe in general); (iv) The Ephemeroptera species involve in general (as well as in taxocenes studied here) both less sensitive “generalists” with beneficial species traits as well as “specialists” with strict environmental limits and poor powers of dispersal that may be easily become extinct; (v) Mayflies have a complex life cycles involving both aquatic and terrestrial phases. Such life cycle type creates evolutionary dichotomy with selection pressure operating in two, more or less, independent environments (cf., e.g. Wilbur 1980 or Brittain 1980, 1990, 1991 and others); (vi) In comparison with other benthic (and aquatic in general) groups of animals their extremely short-lived adult stages represent sole but crucial roles in reproduction, dispersal and recolonization; (vii) Mayflies have survived, despite the problems associated with selection processes operating in both aquatic and terrestrial environments, many climatic shifts and have successfully colonized a very wide range of freshwater habitats from the tropic to the Arctic and from small ponds to large rivers. For instance, in comparison with the

stoneflies (Plecoptera), they have made a greater intrusion into the tropics, both in terms of diversity and abundance being at the same time more abundant and diverse than dragonflies (Odonata) in the Arctic. This fact is extremely important especially with regards to really global climatic changes indications favouring mayflies against at least Plecoptera and Odonata; (viii) Extremely low vagility of “conservative” mayflies with a very low presentation of purely behavioural adaptations most probably reflects the environmental changes much more sensitively than advanced and “progressive” insects groups like, e.g. aquatic dipterans (Diptera) beetles (Coleoptera) and, in a lesser extent also true aquatic bugs (Heteroptera-Nepomorpha) and caddisflies (Trichoptera) and, last but not least, (ix) mayfly species can be relatively easily determined, contrary to the above aquatic insect orders (perhaps except for Heteroptera-Nepomorpha), the larvae of numerous representatives do not provide us with sufficient critical diagnostic characters or even remain undescribed and (x) Principal species traits of numerous mayfly species (or, more precisely, most species living in Central Europe) are relatively well known, despite some gaps of knowledge especiall in rare species and often controversial literature data. This enables, in a relative easy way, to trace crucial species traits with inherent plasticity in response to environmental changes.

There is no doubt that the taxocene composition as well as survival and density of individual species collected mayflies found in the Gránický Brook are influenced, first of all, by their life strategies and reproductive fitness. These attributes (now currently dicussed as “species traits“) in fact represent a certain position of the species in r-K continuum. However, except some exception (cf. Derka, 2003c), r- and/or K-strategies have not still been discussed within the order Ephemeroptera and relative position(s) of species in question remains unknown in detail. Moreover, the situation seems to be rather complicated since, just in the Ephemeroptera, some apparent K-strategists may exhibit a “typical atribute” of r- strategy and *vice versa*. For instance, *Ephemera danica*, a species that might be considered rather K-strategist with regards to some (prevailing) very conservative and fixed attributes (e.g., semivoltinism, no egg laying adaptations, no egg quiescence or survival of drought, larval body form and length, larval growth rate, habitat and substrate range and others) but showing extremely high fecundity, very long winged stage and considerable fligh ability – the attributes considered “typical” for r-strategists. On the other hand, *Baetis rhodani*, an apparent representative of r-stategists as to body form and length, bi- or polyvoltinism, life cycle flexibility, synchrony in egg hatching, larval development and emergence and emergence, pronounced adaptations in oviposition, downstream drift, density and distribution shows very

low fecundity, no respiratory and current speed adaptation and short of winged stage life span– the attributes considered “typical” for K-strategists.

Consequently, we avoid discussing the species found from this point of view since only very limited data concerning the other (here “reference” species) are available and we decided to evaluate species traits or, more precisely, selected traits apparent related to survival and (re-) colonization.

There is a large number of species traits within mayflies, some of them clearly manifested and supported by numerous data in the majority of Central European species (e.g. flexibility of life cycles, fecundity, and length of emergence period), some of them described in detail only in some species (e.g. traits concerning the “ecological range”, see below) and others insufficiently known supported by concrete data only in a very small number of species (e.g. respiratory adaptation, true nature of quiescence or gene flow).

We have selected 25 species traits and tried to classify how they are “advantageous” or “disadvantageous” (cf. Brittain, 1981, 1990, 1991, 2008) for survival under unfavourable conditions. To classify individual species studied we used particular literature data concerning Central European populations species in question (mostly from Clifford, 1982; Landa, 1968, 1969; Landa & Soldán, 1985; Sowa, 1975a, 1975b; Studemann, 1992; Bauernfeind et al., 1995; Haybach, 1998, 2003, 2006; Sartori & Landolt, 1998, 1999; Soldán et al., 1998; Soldán & Zahrádková, 2000; Bauernfeind and Humpesch, 2001; Derka, 2003a, 2003b, 2003c) completed with original unpublished data concerning especially *Electrogena* sp. cf. *ujhelyii* collected at both localities studied and *Ecdyonurus starmachi*, *Rhithrogena carpatoalpina*, and *Paraleptophlebia wernerii* found only in the Klaperův Brook.

To simplify our classification, we have used a simple five grade scale (Tab. 3), describing individual species from most advantageous species trait (++++) to most disadvantageous one (-). In other words, the scale is relative, reflecting only relative intensity (quality) of advantageous species traits, not concrete quantitative data like, e.g., number of eggs produced by a female, percentage of parthenogenetic eggs, growth rate in % of body length increase per day etc. Moreover, the scale is relative also with regards to concrete species classified. For instance, the most advantageous species trend in oviposition is undoubtedly laying of larvae of the first instar (ovoviviparity, and the same is probably valid as far as fecundity is concerned). No such species (at least 2-3 representatives of the genus *Cloeon* in Central Europe apparently ovoviviparous) has been collected at our localities. Consequently, the most advantageous species trait (++++) in this respect is “underwater oviposition” in *Baetis rhodani* and *B. muticus*.

Moreover, there are three more reasons to use only relative (and sometimes rather “intuitive”) classification. First of all, some data are still unknown and could be estimated only by comparison with well known closely related species especially provided that their biology seems to be very similar or even identical. This is the case of *Ecdyonurus starmachi* (scarce data, estimated according to *E. submontanus* and *E. dispar*) and *Paraleptophlebia wernerii* (detailed data limited, estimated according to *P. submarginata* and *P. cincta*). Considerable gaps are apparent especially in species trait called “larval growth rate” – exact growth rates concern mostly related species of the same genus, not species in question.

Further, species traits might be manifested in a different way within different part of species area. For instance, four types of developmental cycle have been identified in *Baetis rhodani*: (a) univoltine seasonal winter cycle (Uw) at latitudes above 65° and in mountains above 900-1,200 m a.s.l., (b) bivoltine seasonal winter life cycle (MBws) or (c) seasonally variable cycle (Uw-MBws) mostly in a central latitudinal belt through Europe and (d) seasonal polyvoltine cycle (MP) in southern area part, two summer generations have been observed, e.g. in the Atlantic Pyrenées. Despite the occurrence of only 1-2 life cycle type within Central Europe in general, (b) and/or (c) (cf. Landa 1968, Sowa 1975b, Studemann et al. 1994, Sartori & Landolt 1998, Bauernfeind & Humpesch 2001), this species shows the highest life cycle flexibility of all remaining species at both localities.

Finally, a lot of literature data seem to be highly controversial. This concerns for example the data on adaptations for oviposition or actual fecundity in *Siphonurus aestivalis*. In the latter case, there are differences from several hundreds to 2-3 thousands of eggs per female and fecundity largely depends on body size, generation, season and different local conditions (see Soldán & Zahrádková 2000 and Bauernfeind & Humpesch 2001 for a complete list of respective references). Nevertheless, the species shows relatively high fecundity however only in our scale comparable to most species of the Heptageniidae and highest fecundity is exhibited by *Ephemera danica* and *Baetis rhodani* while the lowest one by *B. muticus*, quite in concordance with literature data.

Naturally, the significance of individual species traits is rather different. Besides so called “ecological species traits”, e.g. habitat and substrate range, density, distribution, current speed adaptation and eurythermy, we consider so called “biological species traits” more important to explain survival and recolonization (see below). Generally there are three main types of biological traits: (a) those connected with life cycle are the most important and probably determine survival from generation to generation. Besides the type of life cycle (see Material and Methods for their delimitation) this category involves also tendency to polyvoltinism,

flexibility of the cycle length of embryogenesis, egg quiescence, degree of parthenogenesis asynchrony in egg hatch development and emergence and larval growth rate; (b) species traits connected with the adult stage, particularly important as far as the vagility of mayflies is concerned (e.g., fecundity and egg laying adaptation, length of winged stage, flight ability and passive transport); and (c) species traits connected with larval stage and basic physiological functions (e.g. body form and length, feeding type, respiratory adaptations).

Gránický and Klaperův Brook: Comparison of mayfly taxocene survival and (re-)colonization

To evaluate changes in taxocene species composition under unstable hydrological regime contra expected state characteristics of permanent running water of the same type, we compare our study site with the locality on adjacent Klaperův Brook. Judging from respective chemical analyses, the conspicuous impoverishment of taxocene components is definitively not caused by pollution of water and thus it most likely represents a consequence of anomalous hydrological regime that represents a key factor governing the mayfly taxocene composition at least in this case. Moreover, the total absence of pronounced reophilous species indicates that a very important role is played by depth of water body.

Differences in survival at the intermittent Gránický Brook and adjacent permanent Klaperův Brook, which are probably very close to colonization a re-colonization cycles, are expressed by the occurrence of only 4 species in the former while the latter brook is inhabited by additional 9 species (Tab. 2 and Tab. 3), e.g. by rheobionts *Rhithrogena semicolorata* and *R. carpatoalpina* or high oxygen demand species like *Ecdyonurus starmachi*, or semivoltine non-seasonal species showing very plastic developmental cycle like *Ephemera danica* occurring in relatively high densities. The latter species, although exhibits a large number of advantageous species traits (and relatively very high total “plus” score, see Tab. 3) enabling larvae to occur and easily survive at high densities in the Klaperův Brook, is not able to persist in an intermittent brook. Most probably, larval survival is prevented by their relatively high oxygen demands. Contrary to some other species of *Ephemera*, larvae of *E. danica* never occur in lakes, ponds and other standing waters. In addition, traits like substrate range, body form, length, and apparent semivoltinism with no egg dormancy are apparently unfavourable (cf. Brittain, 2008).

Total scores, i.e. number of “pluses” and “minuses” (Table 3), clearly show the species with the highest number of advantageous species traits, in other words, the most successful species in survival and recolonization under unfavourable conditions of intermittent brook, namely *Siphonurus aestivalis* and *Baetis rhodani*. However, both species seems to utilize different

strategies to survive. Survival of the former is enabled mainly by very large ecological range of larvae with apparently low oxygen demands and respiratory adaptation (movable gills). Larvae are known to survive also in periodic water bodies and usually quickly develop in two months in spring during relatively good water supply. Adults fly in May and July. The eggs (hatching usually early next spring with high degree of parthenogenesis, cf. Degrange 1954, 1960) are able to tolerate drying up during summer months (cf. Bohle & Potabgy, 1992 or Fiedler & Bohle, 1994). Sættem & Brittain (1993) even pointed out summer diapause (estivation) in populations living in relatively very warm water. *Siphonurus aestivalis* has not been found in the Klaperův Brook at the places sampled during this study but it occurs, however in very small abundance, at several localities situated upstream studied brook segment.

On the contrary, *Baetis rhodani* seems to be short of the pronounced attributes enabling survival of *S. lacustris* since the larvae are apparently adapted to stream-line habitats. Survival is enabled by oviposition adaptation and mainly enormous flexibility of life cycle accompanied with asynchrony in embryogenesis, larval growth and emergence. There is no doubt that at least some larval cohorts are able to survive partial drying up or re-colonize brook segments that have been subjected to total drought. Concerning other species of the genus *Baetis*, the absence of *B. muticus* in the Gránický Brook is rather difficult to explain. Generally, this species exhibits relatively high number of advantageous species traits, but some of them are apparently less pronounced than those in *B. rhodani*. Since both species always occur at the same places and larvae of *B. muticus* are always less numerous, their competition might be taken into account. Moreover, *B. muticus* generally belongs to the species first disappearing in intermittent brooks or running waters subjected to non periodical water level fluctuation, as seen at numerous localities in the same area (Zahrádková, unpublished).

As to *Electrogena* sp. cf. *ujhelyii*, its life cycle and details on species traits are far from to be understood, published data are extremely scarce. However the total score of species trait in *E.* sp. cf. *ujhelyii* is relatively low in comparison with other survival species, it is, anyway, the highest within the Heptageniidae studied that represent the most sensitive species in general. Larvae with respiratory adaptation (movable gills of most pairs) living among submerged roots or logs at rather slow current places are probably able to survive partial drought. Moreover, they apparently develop in several cohorts and the adults show a pronounced asynchrony in emergence, they fly from April to September.

The occurrence and survival of *Habrophebia fusca* (living in the Gránický Brook) and *H. lauta* (living in the Klaperův Brook) fully agree with the difference in our total score (Tab. 3) as well as with literature data. Larvae of *H. fusca* are evidently much better adapted to survive

unfavourable condition including at least partial drought up, as documented, e.g. in Germany (Fiedler & Bohle, 1994), Spain (Gallardo-Mayenko & Ferreras Romero, 1984) and North Africa (El Agbani et al., 1992). Contrary to *H. lauta*, the species is considered azonal and termophilous (Haybach, 1998, Jacob, 1972), generally tolerating large substrate and habitat range (Landa, 1957, 1969; Sowa, 1975b; Sartori & Landolt, 1999; Derka, 2003b)

Acknowledgements

Research and final preparation of paper was supported by projects No. MSM 0021622416 by Ministry of Education, Youth and Sports of the Czech Republic, No. 206/06/1133 by Grant Agency of the Czech Republic and No. QS500070505 by Grant Agency of the Academy of Sciences of the Czech Republic

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CSN 75 7701 Water quality – Methodology for sampling and treatment of macroinvertebrates from running waters using method PERLA

Figure 1 Location of the study area, the Czech Republic, Europe.



Figure 2 Average daily temperatures, daily precipitations measured at the weather station nearest to the study site (distance 3.5 km) and sampling dates.

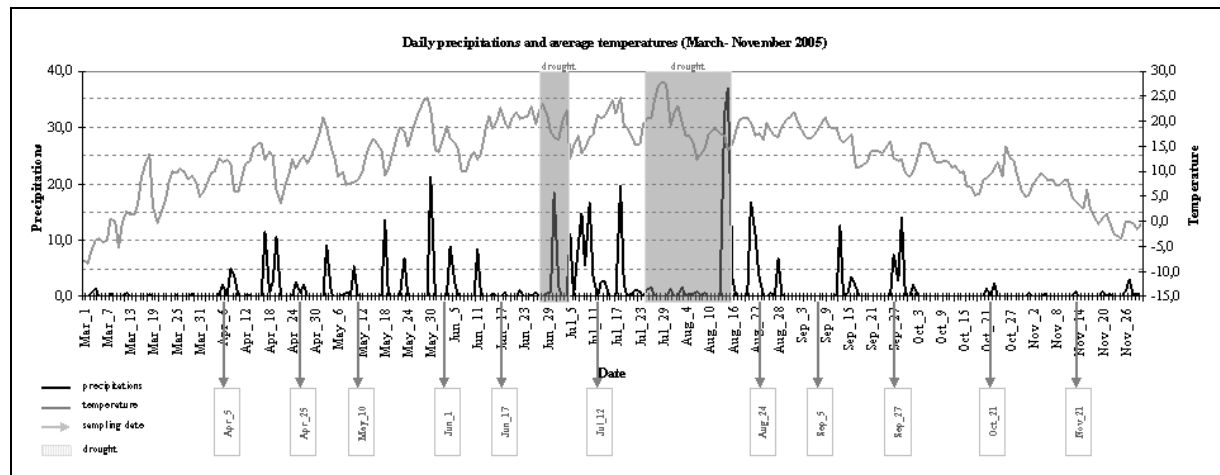


Figure 3 Abundances of mayflies during the year 2005.

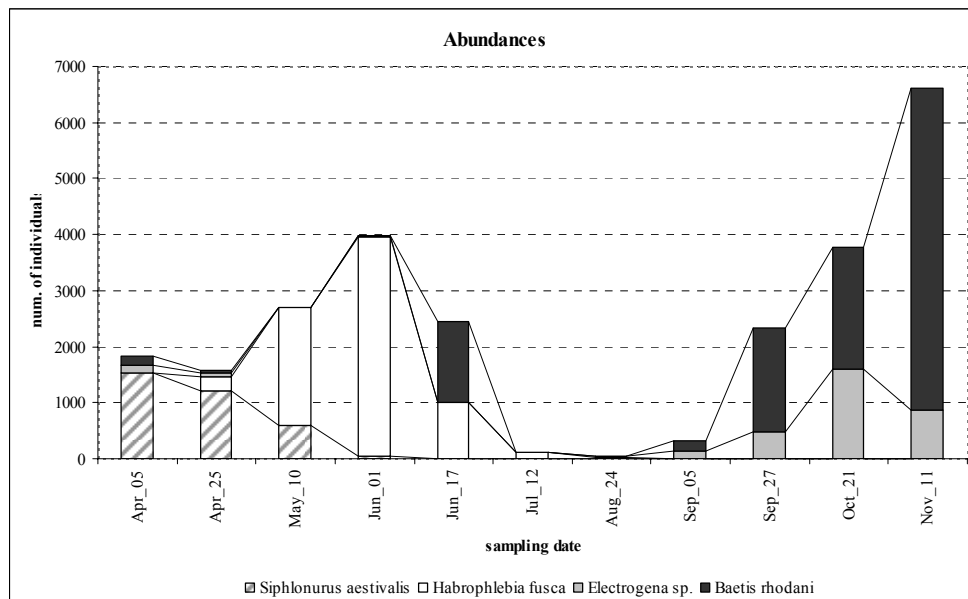


Table 1 Environmental parameters in the Gránický and the Klaperův Brook

	Gránický potok			Klaperův potok		
	min	max	average	min	max	average
water temperature [°C]	6,6	15,2	10,9	3,0	19,0	11,2
pH	6,0	9,0	8,0	6,3	7,9	7,3
conductivity [$\mu\text{S}\cdot\text{cm}^{-1}$]	715	1090	884	594	692	643
dissolved oxygen [$\text{mg}\cdot\text{l}^{-1}$]	7,1	12,1	9,9	8,4	17,4	12,4
depth [cm]	3	30	12	8	15	11
discharge [$\text{m}\cdot\text{s}^{-1}$]	0,001	0,029	0,014	0,130	0,430	0,293

Table 2: Selected traits of mayfly species of the area studied. Feeding types and current preferences according to Schmedtje and Colling (1996); RB (rheobiont), RP (rheophile), RL (rheo- to limnophile); for abbreviations of life cycle types see Material and Methods.

Species	Site	Feeding group					Current preferences	Life cycle type at studied area	Alternative life cycle types
	Gránický Brook Klaperův Brook	Grazers - Scrapers Shredders	Gatherers - Collectors	Passive filter feeders	Active filter feeders				
<i>Siphonurus aestivalis</i> Eaton, 1903	*	1	+	9	0	0	RL	univoltine summer (Us-MBss)	Uw, Landa 1968: A2, B2
<i>Baetis muticus</i> (Linnaeus, 1758)	*	5	0	5	0	0	RP	bivoltine winter-summer(MBws)	Uw, MP
<i>Baetis rhodani</i> (Pictet, 1843)	*	5	0	5	0	0	RP	bivoltine winter-summer (MBws)	Uw, MP
<i>Ecdyonurus starmachi</i> Sowa, 1971	*	7	0	3	0	0	RP	univoltine winter (Uw)	Sowa 1075b: B3
<i>Electrogena</i> sp. cf. <i>ujhelyii</i> (Sowa, 1981)	*	7	0	3	0	0	RP	univoltine winter (Uw)	MBws
<i>Rhithrogena carpatoalpina</i> Klonowska et al., 1987	*	10	0	+	0	0	RB	univoltine winter (Uw)	?MBws, ?Us-Uw, Landa 1968: A1, Sowa 1975b: B2
<i>Rhithrogena semicolorata</i> (Curtis, 1834)	*	10	0	+	0	0	RB	univoltine winter (Uw)	
<i>Habroleptoides confusa</i> Sartori & Jacob, 1986	*	+	0	10	0	0	RP	univoltine winter (Uw)	Landa 1968: A1, Sowa 1975b: B2
<i>Habrophlebia fusca</i> (Curtis, 1834)	*	+	0	10	0	0	RL	univoltine winter (Uw)	Landa 1968: A3, Sowa 1975b: B1
<i>Habrophlebia lauta</i> Eaton, 1884	*	+	0	10	0	0	RL	univoltine winter (Uw)	?Us, Landa 1968: A3, Sowa 1975b: B3
<i>Paraleptophlebia submarginata</i> (Stephens, 1836)	*	+	0	10	0	0	RP	univoltine winter (Uw)	
<i>Paraleptophlebia werneri</i> Ulmer, 1919	*	+	0	10	0	0	RP	univoltine summer (Us)	Landa 1968: A2
<i>Ephemera danica</i> Müller, 1764	*	+	0	0	8	2	RP	semivoltine two years (2Y)	3Y, Uw, Landa 1968: C1, Sowa 1975b: A1

Table 3 Selected species traits of mayfly species found in (*) Gránický Brook and (**) Klaperův Brook, compiled from literature sources and original data (++++: most advantageous, if possible to distinguish with species studied; +++ : highly advantageous or very pronounced; ++ : very advantageous or pronounced; + : advantageous or less pronounced; - : disadvantageous or not pronounced; (?): supposed, not exactly known; (??): literature data highly controversial. Scale +++++ → – defined only within the species collected).

Species found	<i>Siphonurus</i>	<i>Baetis</i>	<i>Baetis</i>	<i>Ecdyonurus</i>	<i>Electrogena</i>	<i>Rhithrogena</i>	<i>Rhithrogena</i>	<i>Habroleptoides</i>	<i>Habrophlebia</i>	<i>Habrophlebia</i>	<i>P.</i>	<i>P.</i>	<i>E.danica</i>
(*) - Gránický Brook	<i>aestivalis</i>	<i>muticus</i>	<i>rhodani</i>	<i>starmachi</i>	<i>sp.cf. ujhehyii</i>	<i>carpatoalpina</i>	<i>semicolorata</i>	<i>confusa</i>	<i>fusca</i>	<i>lauta</i>	<i>submarginata</i>	<i>weneri</i>	(*)
(**) - Klaperův Brook		(*)	(*)	(*)	(*)	(*)	(*)	(*)		(*)	(*)	(*)	
Selected advantageous species traits and scores	(**)		(**)		(**)				(**)				
Life cycle flexibility	+++	+++	++++	–	–	–	+	–	+(?)	+(?)	–	–	+++
Bi- or polyvoltinism	+++	++++	++++	–	–	–	+(?)	–	–	–	–	–	–
Fecundity	++(??)	+	+++	++	+	++	++	+	+	+	+	+(?)	++++
Adaptation in oviposition	–(??)	++++	++++	+++(?)	+++(?)	+++	+++	+++	+++(?)	+++(?)	+++(?)	+++(?)	+(??)
Length of embryogenesis	++++	++	++	–	+(?)	–	–	+	++	++	+	+++	+
Egg quiescence	++++	+	+	–	–	–	–	–	–	–	–	+++ (?)	–
Egg survival of drought	++++	+(?)	++	–	+(?)	–	–	+	+++(?)	++	+	+(?)	+
Degree of parthenogenesis	++++	++	++	+(?)	+(?)	+(?)	+	+	+	+	+	+(?)	++
Asynchrony in egg hatch	++	+++	++++	+	++	+	+	++	++	++	+	+	+
Asynchronic development	++	+++	++++	+	++(?)	–	+	++	++	++	+	–	+++
Asynchrony in emergence	+++	++++	++++	++	++	++	++	+++	++	++	++	+	+++
Length of winged stage	+++	+	+	+++	+++	+++	+++	++	++	++	+++	++	++++
Flight ability	+++	+	+	+++	+++	+++	+++	++	++	++	++	++	++++

Passive transport	++	++++	++++	++	++	++	++	+++	+++	+++	++	++	+
Body form and length	++	++++	++++	–	–	–	–	+	++	++	++	+	– (??)
Larval growth rate	+++ (?)	+++ (?)	++++	++ (?)	++ (?)	++	++	+	+(?)	+(?)	+++	+++ (?)	–
Eurythermy	++	+	+++	–		–	–	++	+++	++	+	++	++
Respiratory adaptation	++++	–	–	+	+	–	–	++	++	++	++	++	+++
Feeding type	++++	+++	+++	+	++	+	+	++	++	++	++	++	+
Current speed adaptation	++++	+	+	–	+	–	–	++	+++	++	++	++++	++
Habitat range	++++	++	+++	–	+	–	–	++	+++	++	++	+++	++
Substrate range	++++	++	++	+	++	+	+	+++	+++	+++	++	++	+
Downstream drift	++	++++	++++	++	++ (?)	+++	+++	+	+	+	+	– (?)	+
Density	+++	+++	++++	++	+	+++	+++	++	++	+	++	+	+++
Distribution	+++	++++	++++	++	++	++	++	++	+++	+++	+++	+	+++
Score of most (4+) highly (3+) and very (2+) advantageous species trait	9 (4+)	7 (4+)	12 (4+)	0 (4+)	0 (4+)	0 (4+)	0 (4+)	0 (4+)	0 (4+)	0 (4+)	0 (4+)	1 (4+)	3 (4+)
	8 (3+)	6 (3+)	4 (3+)	3 (3+)	3 (3+)	5 (3+)	5 (3+)	4 (3+)	8 (3+)	4 (3+)	4 (3+)	5 (3+)	6 (3+)
	7 (2+)	4 (2+)	4 (2+)	7 (2+)	9 (2+)	5 (2+)	5 (2+)	11 (2+)	10 (2+)	13 (2+)	10 (2+)	7 (2+)	4 (2+)
Total score, No. of (+) and (–)	74 (+)	61 (+)	72 (+)	29 (+)	35 (+)	29 (+)	32 (+)	41 (+)	49 (+)	44 (+)	40 (+)	41 (+)	46 (+)
	1 (–)	1 (–)	1 (–)	9 (–)	4 (–)	11 (–)	8 (–)	3 (–)	2 (–)	2 (–)	3 (–)	4 (–)	4 (–)